TYPICAL y PARAMETERS FOR TYPES 40822 and 40823

y Parameters vs. V_{DS}

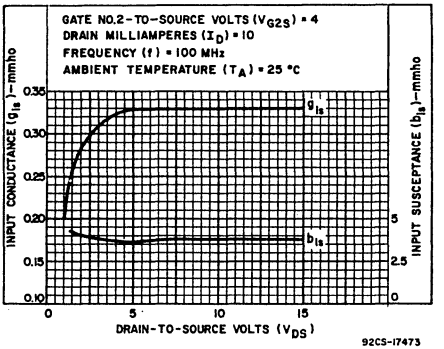


Fig. 8 - y_{is} vs. V_{DS}

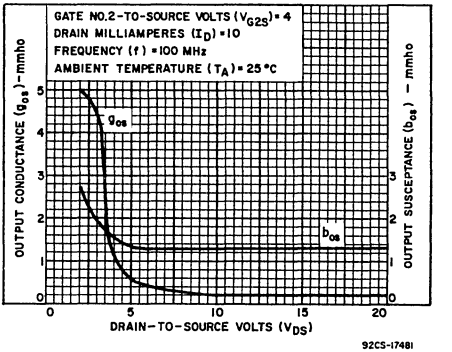


Fig. 9 - y_{os} vs. V_{DS}

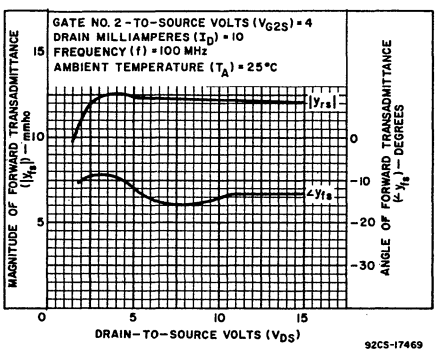


Fig. 10 - y_{fs} vs. V_{DS}

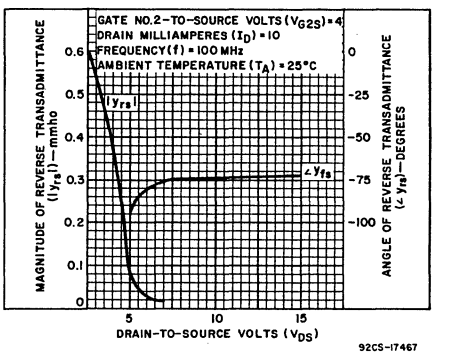


Fig. 11 - y_{rs} vs. V_{DS}

y Parameters vs. I_D

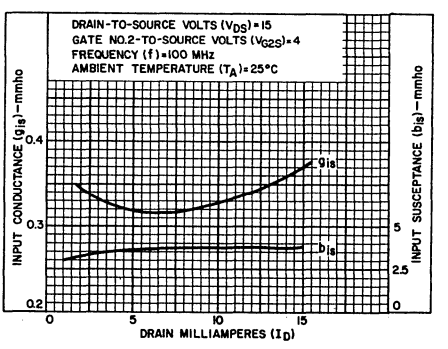


Fig. 12 - y_{is} vs. I_D

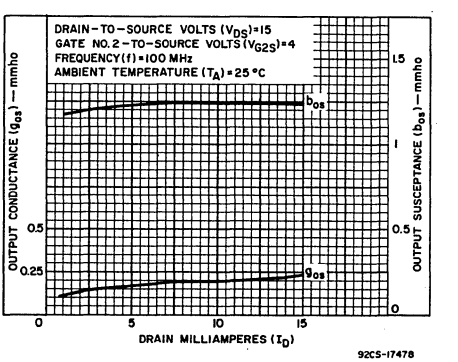


Fig. 13 - y_{os} vs. I_D

TYPICAL y PARAMETERS FOR TYPES 40822 and 40823

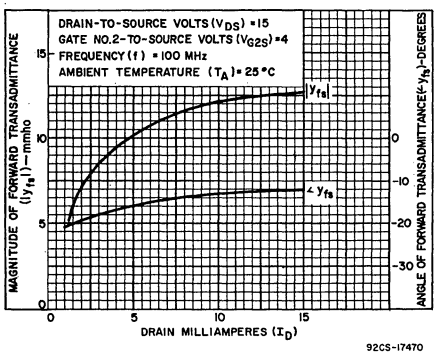


Fig. 14 - y_{fs} vs. I_D

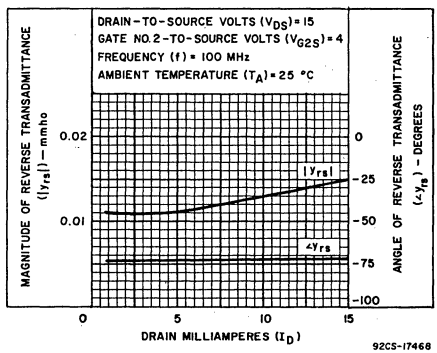


Fig. 15 - y_{rs} vs. I_D

y Parameters vs. V_{G2S}

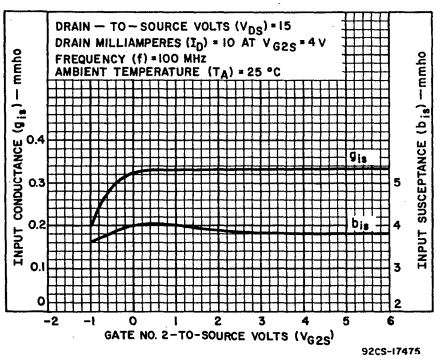


Fig. 16 - y_{is} vs. V_{G2S}

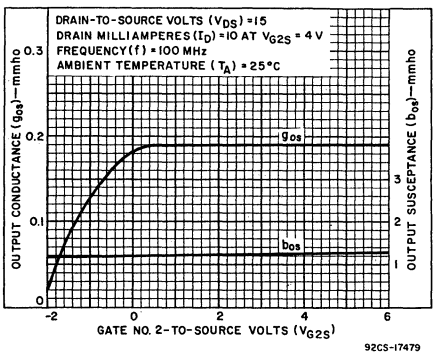


Fig. 17 - y_{os} vs. V_{G2S}

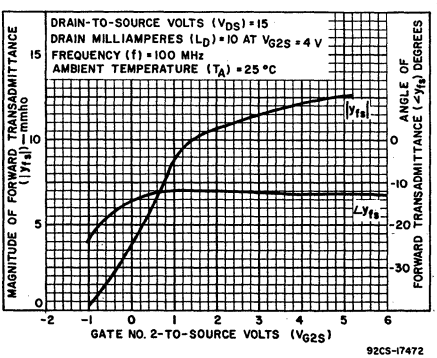


Fig. 18 - y_{fs} vs. V_{G2S}

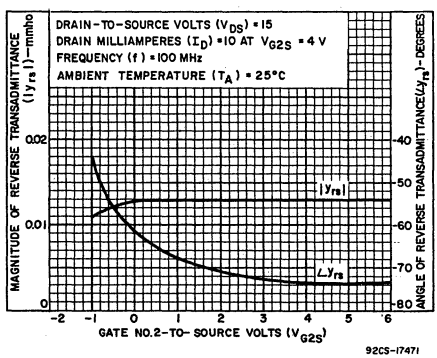


Fig. 19 - y_{rs} vs. V_{G2S}

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Types

40841



JEDEC TO-72 H-1299

Silicon Dual-Insulated Gate Field-Effect Transistor

With Integrated Gate-Protection Circuits

General-Purpose Economy Type for Applications from DC to 500 MHz

Applications:

- DC amplifiers
- RF amplifiers
- mixers
- IF amplifiers
- video amplifiers
- differential amplifiers
- frequency multipliers
- choppers
- voltage-controlled attenuators
- constant-current source
- voltage regulators
- telemetry & multiplex
- servo amplifiers
- proximity switches
- phase splitters
- thyristor trigger circuits
- industrial timers — long time delays

RCA-40841* is an n-channel silicon, depletion type, dual-insulated gate, field-effect transistor intended for general-purpose applications from DC to frequencies up to 500 MHz.

This MOS/FET provides excellent power gain, linear-circuit operation and has a wide dynamic operating range. Its square-law characteristics result in low cross-modulation performance over the AGC range. Its dual-gate construction reduces feedback capacitance by shielding Gate No. 1 from the drain, and makes it possible to isolate the local oscillator signal from the incoming signal bv applying the two signals to separate gates. The very low feedback capacitance of this device eliminates the need for neutralization in circuits using the dual-gate configuration. Use of the device in the RF input stage of a receiver reduces local oscillator feed-through to the antenna. The 40841 requires negligible AGC power, provides automatic delay when AGC is applied to Gate No. 2, and exhibits slight input impedance variations during AGC functioning. The device has exceptionally high input impedance, an attribute for timing-circuit design.

Back-to-back diodes are fabricated on the same monolithic silicon pellet as the MOS/FET to protect the gates against damage due to electrostatic charges frequently encountered during normal handling. These back-to-back diodes also function as "transient trappers" by limiting in-circuit transient voltages that exceed ±10 volts.

Maximum ratings and electrical characteristics are included in the data for operation of the 40841 as the equivalent of a single-gate device. For single-gate operation, connect Gate No. 1 (Term. 2) to Gate No. 2 (Term. 3), as shown in the Terminal Diagrams on Page 2. The 40841 MOS/FET is hermetically sealed in the metal JEDEC TO-72 package.

The following dual-gate MOS/FET types are specified for applications requiring premimum-grade performance: 3N200, 3N187, 40673, 40819, 40820, 40821, 40822, and 40823.

Device Features:

- back-to-back diodes protect gate insulation against damage due to static changes frequently encountered during handling
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high power gain: G_{ps} = 32 dB (typ.) at 44 MHz
- gate leakage currents: I_{G1SS} and I_{G2SS} = 60 nA (max.) at T_A = 25°C
- high input impedance
- excellent thermal stability

Performance Features:

- superior cross-modulation performance and greater dynamic range than bipolar and junction-gate FETs
- wide dynamic range permits large-signal handling before overloading
- virtually no agc power required
- greatly reduced spurious responses in AM and FM receivers
- dual-gate configuration permits simplified AGC circuitry
- operates at frequencies to 500 MHz without neutralization in circuits utilizing the dual-gate configuration
- operates up to UHF with low-noise performance

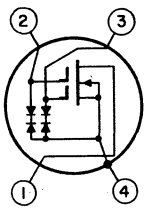
Detailed information, utilizing RCA dual-gate protected MOS/FETs in RF applications, is given in the following RCA Application Notes: AN-4431 "RF Applications of the Dual-Gate MOS/FET up to 500 MHz" and AN-4018 "Design of Gate-Protected MOS Field-Effect Transistors".

* Formerly Developmental Type TA8242.

Maximum Ratings	Dual-Gate Configuration	Single-Gate Configuration	
Absolute Maximum Values, at T_A = 25°C:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +18	−0.2 to +18	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	−	μA
Gate Terminal Current, I_{GS}	−	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+24	−	V
Drain-to-Gate Voltage, V_{DG}	−	+24	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C
Continuous Working Voltage#, at T_A = 25°C:			
Gate No. 1-to-Source Voltage, V_{G1S}	−4.5 to +3	−	V
Gate No. 2-to-Source Voltage, V_{G2S}	-4.5 to +4.5 or 40% of V_{DS} (whichever value is less)	−	V
Gate-to-Source Voltage, V_{GS}	−	−4.5 to +3	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	−	V
Drain-to-Gate Voltage, V_{DG}	−	+20	V

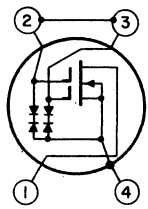
#Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the Absolute Maximum Ratings are not exceeded.

TERMINAL DIAGRAMS



DUAL-GATE CONFIGURATION

LEAD 1—DRAIN
LEAD 2—GATE No.2
LEAD 3—GATE No.1
LEAD 4—SOURCE
SUBSTRATE AND CASE



SINGLE-GATE CONFIGURATION

LEAD 1—DRAIN
LEADS—2 AND 3—GATE
LEAD 4—SOURCE,
SUBSTRATE AND CASE

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS CONFIGURATION DUAL-GATE MIN.	DUAL-GATE TYP.	DUAL-GATE MAX.	SINGLE-GATE MIN.	SINGLE-GATE TYP.	SINGLE-GATE MAX.	UNITS
Gate-to-Source Cutoff Voltage:										
Dual-Gate (No. 1)	$V_{G1S(off)}$	V_{DS} = +15V, I_D=200μA, V_{G2S} = +4V		–	–2	–	–	–	–	V
Dual-Gate (No. 2)	$V_{G2S(off)}$	V_{DS} = +15V, I_D=200μA, V_{G1S}=0		–	–2	–	–	–	–	V
Single Gate	$V_{GS(off)}$	V_{DS} = +15V, I_D=200μA		–	–	–	–	–1.6	–	V
Gate-to-Source Forward Breakdown Voltage:		I_{G1SSF} =								
Dual-Gate (No. 1)	$V_{(BR)G1SSF}$	I_{G2SSF} =	$V_{G2S} = V_{DS} = 0$	–	9	–	–	–	–	V
Dual-Gate (No. 2)	$V_{(BR)G2SSF}$	100μA	$V_{G1S} = V_{DS} = 0$	–	9	–	–	–	–	V
Single-Gate	$V_{(BR)GSSF}$	I_{GSSF} = 100μA, V_{DS} = 0		–	–	–	–	9	–	V
Gate-to-Source Reverse Breakdown Voltage:		I_{G1SSR} =								
Dual-Gate (No. 1)	$V_{(BR)G1SSR}$	I_{G2SSR} =	$V_{G2S} = V_{DS} = 0$	–	9	–	–	–	–	V
Dual-Gate (No. 2)	$V_{(BR)G2SSR}$	100μA	$V_{G1S} = V_{DS} = 0$	–	9	–	–	–	–	V
Single-Gate	$V_{(BR)GSSR}$	I_{GSSR} = 100μA, V_{DS} = 0		–	–	–	–	9	–	V
Gate Terminal Forward Current:										
Dual-Gate (No. 1)	I_{G1SSF}	$V_{DS} = V_{G2S} = 0$, V_{G1S} = 6 V		–	–	60	–	–	–	nA
Dual Gate (No. 2)	I_{G2SSF}	$V_{DS} = V_{G1S} = 0$, V_{G2S} = 6 V		–	–	60	–	–	–	nA
Single-Gate	I_{GSSF}	V_{DS} = 0, V_{GS} = 6 V		–	–	–	–	–	120	nA
Gate Terminal Reverse Current:										
Dual-Gate (No. 1)	I_{G1SSR}	$V_{DS} = V_{G2S} = 0$, V_{G1S} = –6 V		–	–	60	–	–	–	nA
Dual-Gate (No. 2)	I_{G2SSR}	$V_{DS} = V_{G1S} = 0$, V_{G2S} = –6 V		–	–	60	–	–	–	nA
Single-Gate	I_{GSSR}	V_{DS} = 0, V_{GS} = –6 V		–	–	–	–	–	120	nA
Zero-Bias Drain Current:										
Dual-Gate	I_{DS}	V_{DS} = +15 V, V_{G1S} = 0, V_{G2S} = +4 V		–	10	–	–	–	–	mA
Single-Gate	I_{DSS}	V_{DS} = +15 V, V_{GS} = 0		–	–	–	–	3.7	–	mA
Forward Transconductance (Gate-to-Drain)										
Dual-Gate	g_{fs}		1 kHz	–	12000	–	–	–	–	μmho
Single-Gate	g_{fs}			–	–	–	–	7000	–	μmho
Small-Signal, Short-Circuit Input Capacitance†	C_{iss}		f = 1 MHz	–	6.5	–	–	11	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦	C_{rss}	V_{DS} = +15 V, I_D = 10 mA, [Dual-Gate only V_{G2S} = +4 V]		–	0.02	–	–	0.54	–	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}			–	2	–	–	2	–	pF
Audio Spot Noise Figure*										
Dual-Gate	NF		f = 1 kHz	–	0.46	–	–	–	–	dB
Single-Gate	NF		f = 1 kHz	–	–	–	–	0.29	–	dB
Power Gain	G_{ps}		44 MHz	–	32	–	–	–	–	dB
Conversion Gain	$G_{ps(C)}$			–	24	–	–	–	–	dB

† Capacitance between Gate No. 1 and all other terminals (Dual-Gate), Gate and all other terminals (Single-Gate)

♦ Three-terminal measurement with Gate No. 2 and Source returned to guard terminal (Dual-Gate)

* Noise Figure = $10 \log_{10}\left[1 + \frac{e_n^2}{4\,KT\,BW\,R_g}\right]$ where K = 1.38×10^{-23}; T = Temperature in °Kelvin; BW = Bandwidth in Hz; R_g = Generator resistance

TYPICAL CHARACTERISTICS FOR 40841 IN DUAL-GATE CONFIGURATION

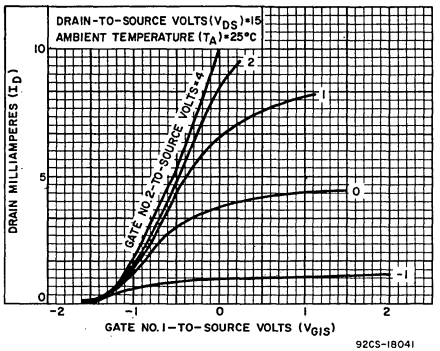


Fig.1–I_D vs. V_{G1S}.

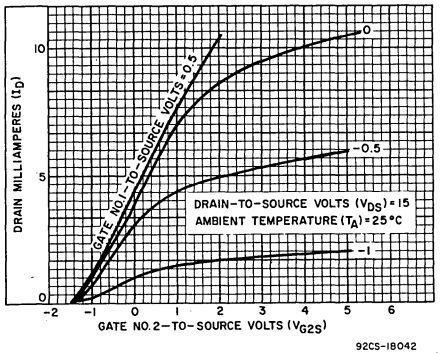


Fig.2–I_D vs. V_{G2S}.

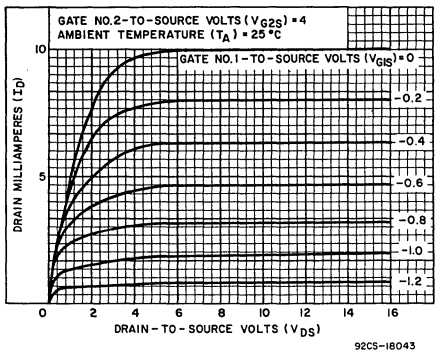


Fig.3–I_D vs. V_{DS}.

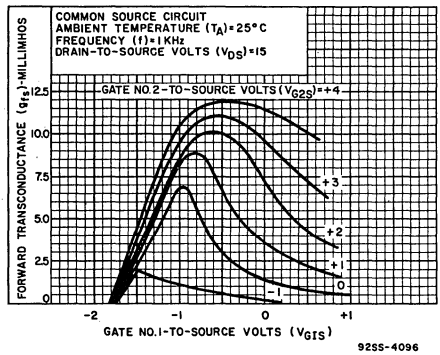


Fig.4–g_{fs} vs. V_{G1S}.

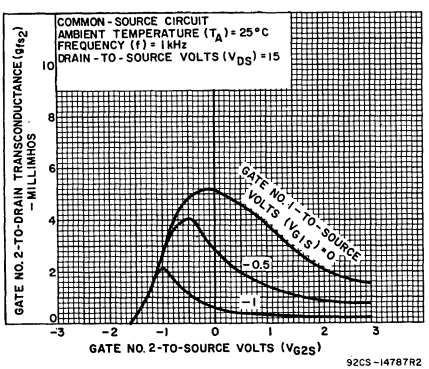


Fig.5–g_{fs2} vs. V_{G2S}.

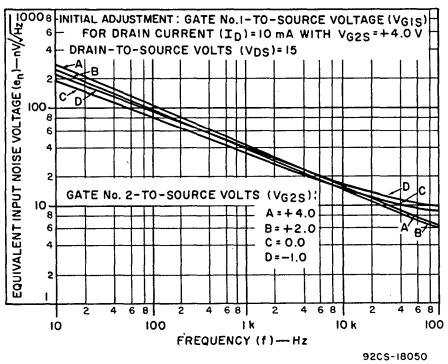


Fig.6–e_n vs. f.

TYPICAL CHARACTERISTICS FOR 40841 IN SINGLE-GATE CONFIGURATION
(Terminals 2 and 3 tied together to comprise effective single-gate)

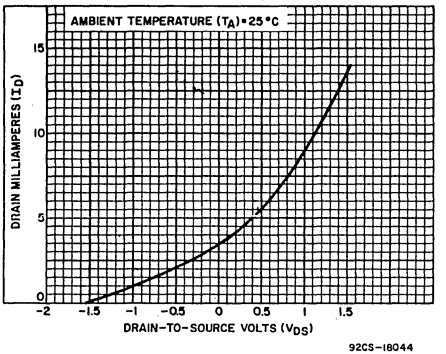


Fig.7–I_D vs. V_{DS}.

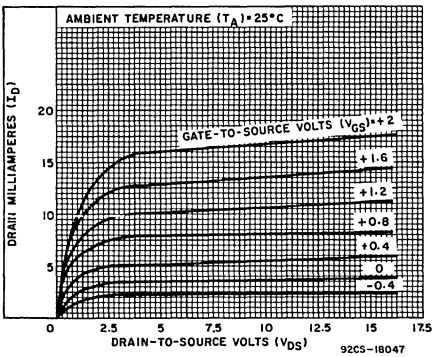


Fig.8–I_D vs. V_{DS}.

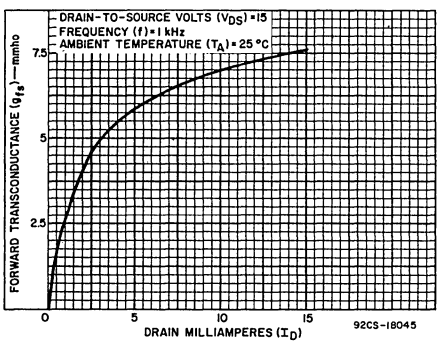


Fig.9–g_{fs} vs. I_D.

TYPICAL CHARACTERISTICS FOR 40841 IN SINGLE-GATE & DUAL-GATE CONFIGURATION

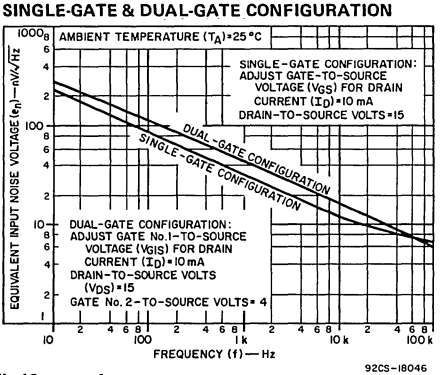


Fig.10–e_n vs. f.

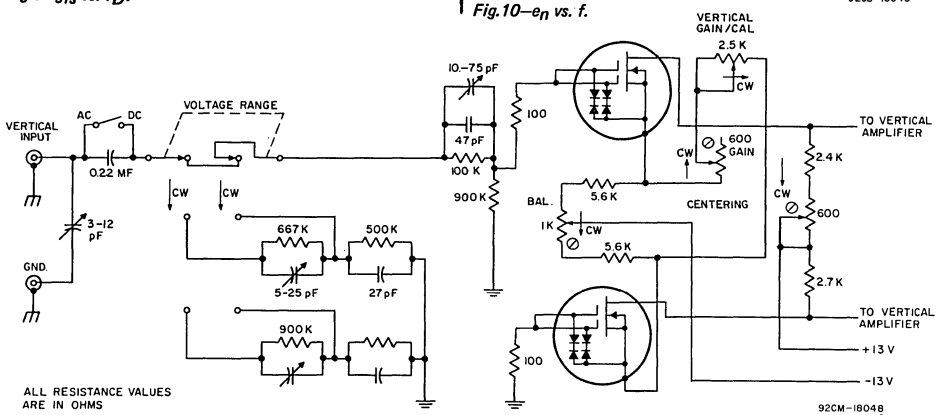
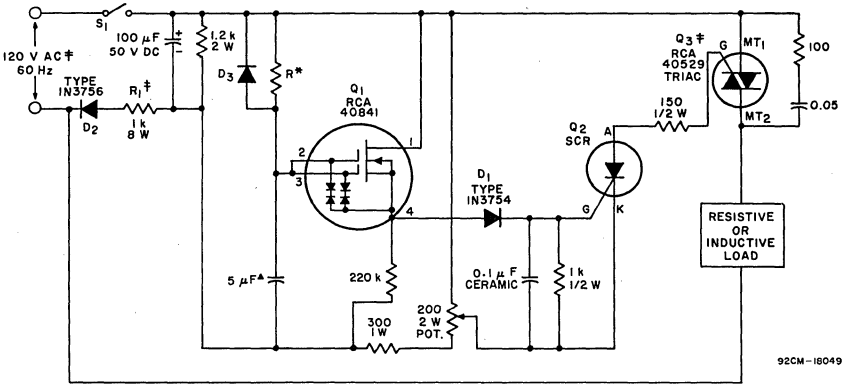


Fig.11–Typical differential amplifier utilizing the 40841 in the vertical input stage of a solid-state oscilloscope.

SOLID-STATE TIMER FOR INDUSTRIAL APPLICATIONS



▲ Cornell-Dubilier Electronics—Type MMW or equivalent.

* R controls duration of time delay. At R = 60 MΩ up to 5-minute delay (IRC resistor, Type CGH or equivalent)

‡ This circuit can also be used at supply voltages of 240 V AC and 24V AC (60Hz) by changing the values of R1 and Q3.

TIMING CIRCUIT CHARACTERISTICS

T_A = −25°C to +60°C

Accuracy: ±10% (over temperature)

Repeatability: ±3% (at 25°C)

Reset Time: Less than 150 ms

Q2: V_{DRM} = 60V, I_{GT} = 200μA, I_T = 0.8A

D3: I_R = 1 nA, V_R = 60V

Fig.12—Typical timing circuit utilizing the 40841 in a single-gate configuration.

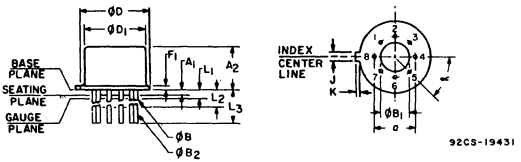
OPERATING CONSIDERATIONS

The flexible leads of the 40841 are usually soldered to the circuit elements. As in the case with any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.

Dimensional Outlines

DIMENSIONAL OUTLINES FOR LINEAR INTEGRATED CIRCUITS AND MOS/FET'S

JEDEC MO-002-AL 8-LEAD TO-5 STYLE

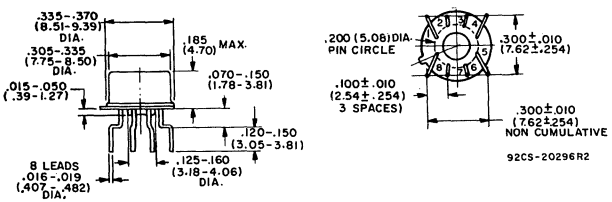


SYMBOL	INCHES MIN.	INCHES MAX.	NOTE	MILLIMETERS MIN.	MILLIMETERS MAX.
a	0.200 TP		2	5.88 TP	
A_1	0.010	0.050		0.26	1.27
A_2	0.165	0.185		4.20	4.69
øB	0.016	0.019	3	0.407	0.482
$øB_1$	0.125	0.160		3.18	4.06
$øB_2$	0.016	0.021	3	0.407	0.533
øD	0.335	0.370		8.51	9.39
$øD_1$	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
j	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
α	45° TP			45° TP	
N	8		6	8	
N_1	3		5	3	

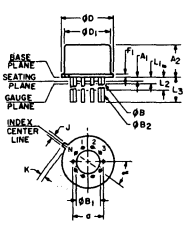
NOTES

1. Refer to Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178mm) radius of True Position (TP) at maximum material condition.
3. øB applies between L_1 and L_2. $øB_2$ applies between L_2 and 0.500" (12.70mm) from seating plane. Diameter is uncontrolled in L_1 and beyond 0.500" (12.70mm).
4. Measure from Max. øD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

8-LEAD TO-5 STYLE WITH DUAL-IN-LINE FORMED LEADS (DIL-CAN)



10-LEAD TO-5 STYLE PACKAGE JEDEC MO-006-AF

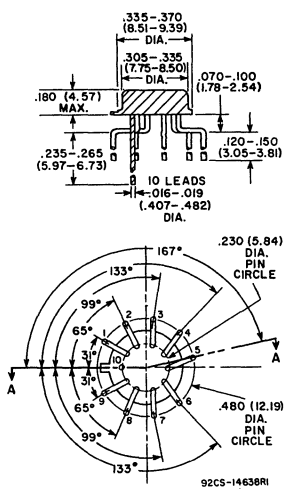


SYMBOL	INCHES MIN.	INCHES MAX.	NOTE	MILLIMETERS MIN.	MILLIMETERS MAX.
a	0.230 TP		2	5.84 TP	
A_1	0	0		0	0
A_2	0.165	0.185		4.19	4.70
øB	0.016	0.019	3	0.407	0.482
$øB_1$	0	0		0	0
$øB_2$	0.016	0.021	3	0.407	0.533
øD	0.335	0.370		8.51	9.39
$øD_1$	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
j	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
α	36° TP			36° TP	
N	10		6	10	
N_1	1		5	1	

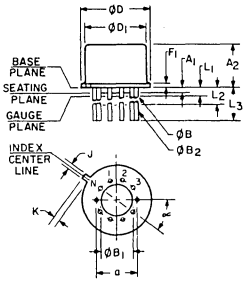
NOTES:

1. Refer to Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178 mm) radius of True Position (TP) at maximum material condition.
3. øB applies between L_1 and L_2. $øB_2$ applies between L_2 and 0.500" (12.70 mm) from seating plane. Diamerer is uncontrolled in L_1 and beyond 0.500" (12.70 mm).
4. Measure from Max. øD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

10 FORMED LEADS RADIALLY ARRANGED TO-5 TYPE



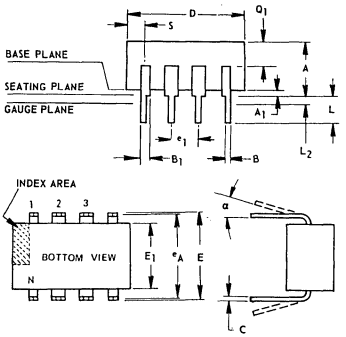
12-LEAD TO-5 STYLE PACKAGE JEDEC MO-006-AG



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
a	0.230		2	5.84 TP	
A_1	0	0		0	0
A_2	0.165	0.185		4.19	4.70
øB	0.016	0.019	3	0.407	0.482
$øB_1$	0	0		0	0
$øB_2$	0.016	0.021	3	0.407	0.533
øD	0.335	0.370		8.51	9.39
$øD_1$	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
j	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
	30 TP			30 TP	
N	12		6	12	
N_1	1		5	1	

NOTES:
1. Refer to Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178 mm) radius of True Position (TP) at maximum material condition.
3. øB applies between L_1 and L_2. $øB_2$ applies between L_2 and 0.500" (12.70 mm) from seating plane. Diamerer is uncontrolled in L_1 and beyond 0.500" (12.70 mm).
4. Measure from Max. øD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

8-LEAD DUAL-IN-LINE PLASTIC PACKAGE (MINI-DIP)



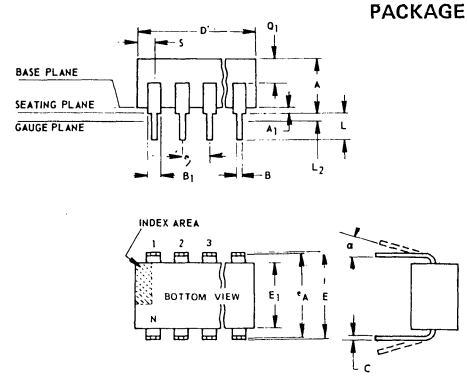
SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.508	1.27
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.889	1.65
C	0.008	0.012*		0.203	0.304
D	0.370	0.400		9.40	10.16
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.762
α	0°	15°	4	0°	15°
N	8		5	8	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.015	0.060		0.381	1.52

NOTES:
1. Refer to Rules for Dimensioning (JEDEC Publication No. 13) for Axial Lead Product Outlines.
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at gauge plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation.
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

* When device is supplied solder dipped, the maximum lead thickness (narrow portion) will not exceed 0.013".

92CS-24026

8-LEAD DUAL-IN-LINE FRIT-SEAL PACKAGE

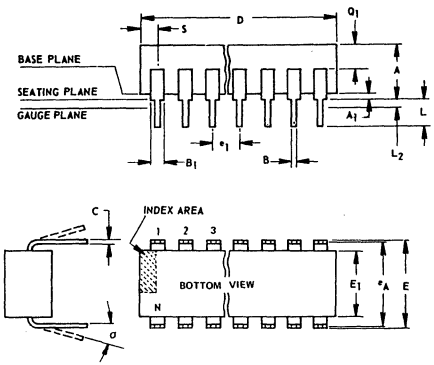


SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.508	1.27
B	0.014	0.020		0.356	0.508
B_1	0.050	0.065		1.27	1.65
C	0.008	0.012		0.203	0.304
D	0.376	0.396		9.55	10.05
E	0.315	0.345		8.00	8.76
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.100	0.150		2.54	3.81
L_2	0.000	0.030		0.000	0.762
α	0°	15°	4	0°	15°
N	8		5	8	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.020	0.060		0.508	1.52

92CM-20827

NOTES:
1. Refer to Rules for Dimensioning (JEDEC Publication No. 13) for Axial Lead Product Outlines.
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at guage plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

14-LEAD DUAL-IN-LINE PLASTIC AND FRIT-SEAL PACKAGE
JEDEC MO-001-AB



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.51	1.27
B	0.014	0.020		0.356	0.508
B_1	0.050	0.065		1.27	1.65
C	0.008	●0.012		0.204	0.304
D	0.745	0.770		18.93	19.55
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	14		5	14	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.065	0.090		1.66	2.28

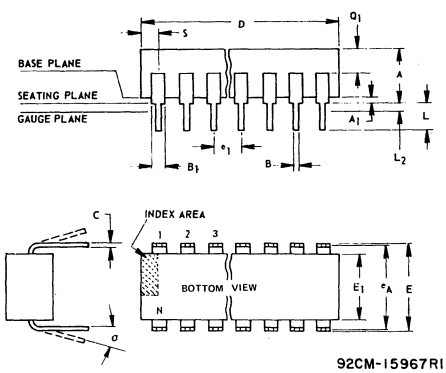
92SS-4296RI

NOTES:

1. Refer to Rules for Dimensioning (JEDEC Publication No. 13) for Axial Lead Product Outlines.
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at guage plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation.
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

● When this device is supplied solder-dipped, the maximum lead thickness (narrow portion) will not exceed 0.013".

16-LEAD DUAL-IN-LINE PLASTIC AND FRIT- SEAL PACKAGE
JEDEC MO-001-AC



92CM-15967RI

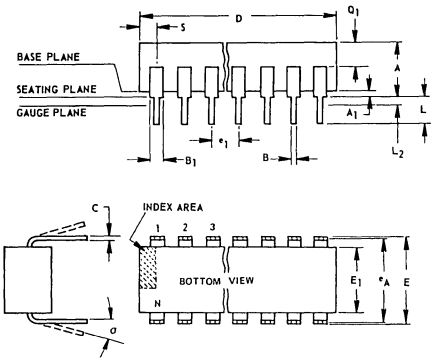
SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.51	1.27
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.89	1.65
C	0.008	●0.012		0.204	0.304
D	0.745	0.785		18.93	19.93
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	16		5	16	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.015	0.060		0.39	1.52

NOTES:

1. Refer to Rules for Dimensioning (JEDEC Publication No. 13) for Axial Lead Product Outlines.
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at gauge plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation.
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

● When this device is supplied solder-dipped, the maximum lead thickness (narrow portion) will not exceed 0.013".

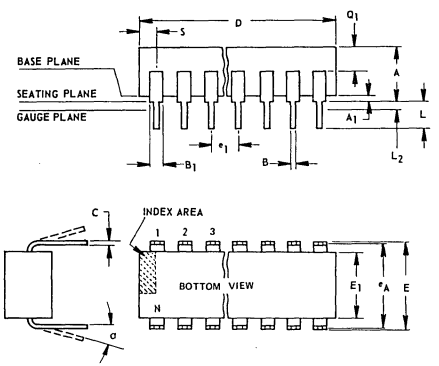
14-LEAD DUAL-IN-LINE CERAMIC PACKAGE JEDEC MO-001-AD



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.120	0.160		3.05	4.06
A_1	0.020	0.065		0.51	1.65
B	0.014	0.020		0.356	0.508
B_1	0.050	0.065		1.27	1.65
C	0.008	●0.012		0.204	0.304
D	0.745	0.770		18.93	19.55
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	14		5	14	
N_1	0		6	0	
Q_1	0.050	0.085		1.27	2.15
S	0.065	0.090		1.66	2.28

92SS-44IIRI

16-LEAD DUAL-IN-LINE CERAMIC PACKAGE JEDEC MO-001-AE



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.120	0.160		3.05	4.06
A_1	0.020	0.065		0.51	1.65
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.89	1.65
C	0.008	●0.012		0.204	0.304
D	0.745	0.785		18.93	19.93
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	16		5	16	
N_1	0		6	0	
Q_1	0.050	0.085		1.27	2.15
S	0.015	0.060		0.39	1.52

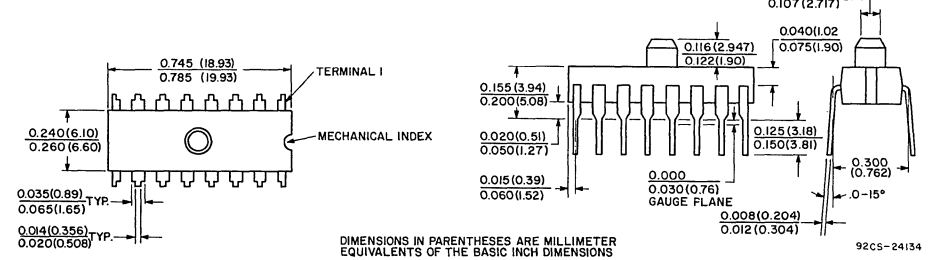
92SS-4286R3

Notes for 14- and 16-Lead Dual-in-Line Ceramic Packages

1. Refer to Rules for Dimensioning (JEDEC Publication No. 13) for Axial Lead Product Outlines.
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at guage plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation.
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

●When this device is supplied solder-dipped, the maximum lead thickness (narrow portion) will not exceed 0.013".

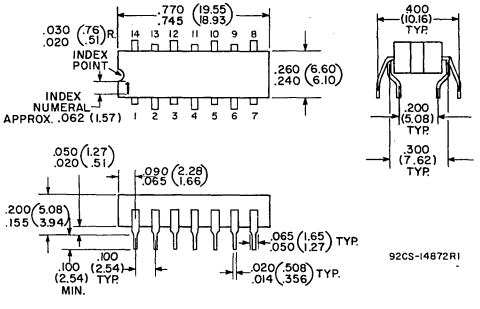
16-LEAD DUAL-IN-LINE "POWER-STUD" PLASTIC PACKAGE

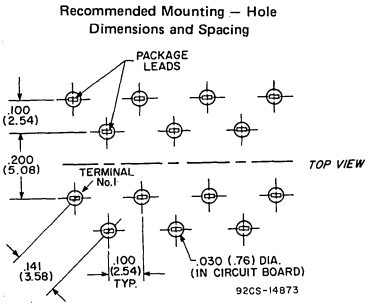


DIMENSIONS IN PARENTHESES ARE MILLIMETER EQUIVALENTS OF THE BASIC INCH DIMENSIONS

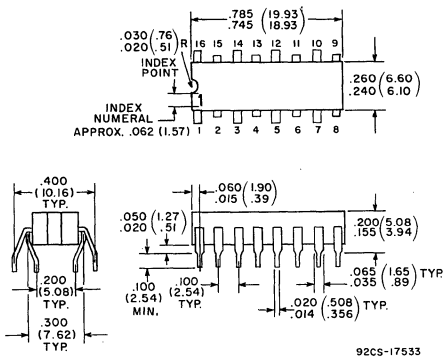
92CS-24134

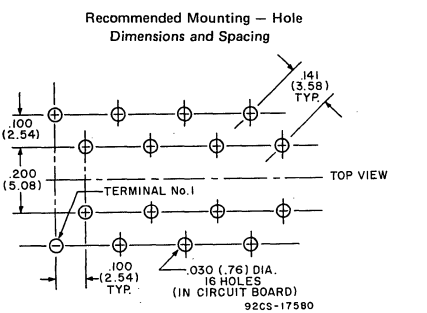
14-LEAD QUAD-IN-LINE PLASTIC PACKAGE



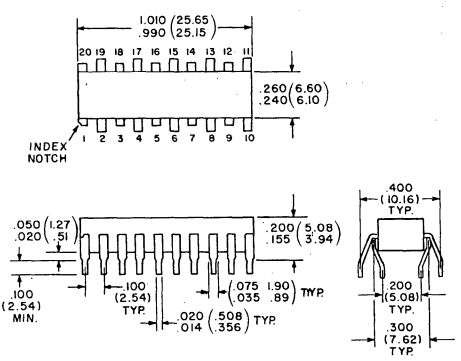


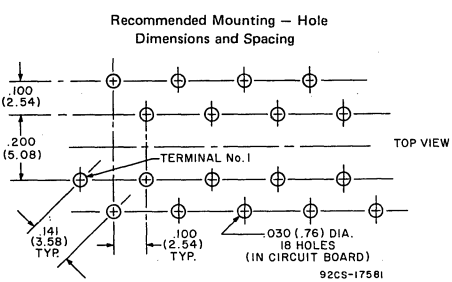
16-LEAD QUAD-IN-LINE PLASTIC PACKAGE



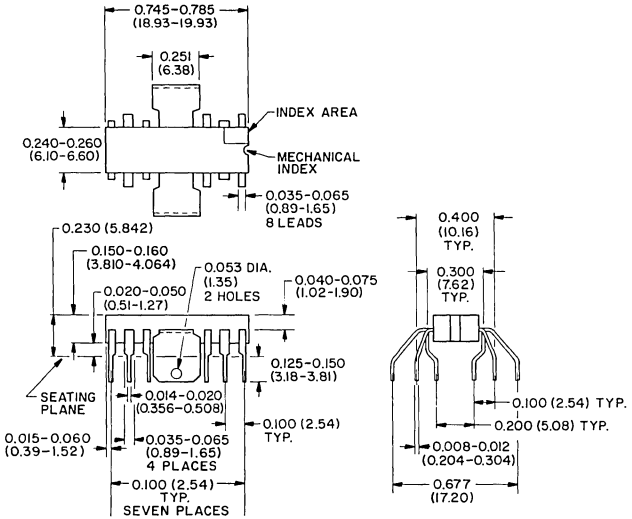


20-LEAD QUAD-IN-LINE PLASTIC PACKAGE





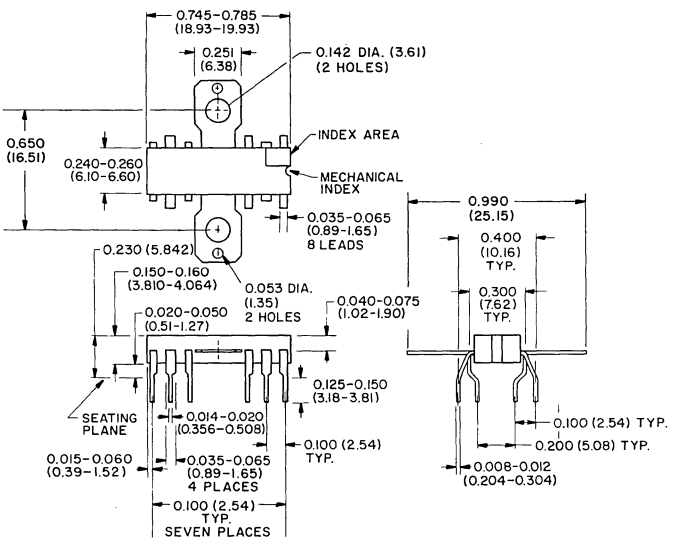
MODIFIED 16-LEAD QUAD-IN-LINE PLASTIC PACKAGE WITH INTEGRAL BENT DOWN WING-TAB HEAT SINK



DIMENSIONS IN PARENTHESES ARE MILLIMETER EQUIVALENTS OF THE BASIC INCH DIMENSIONS

92CM-25044

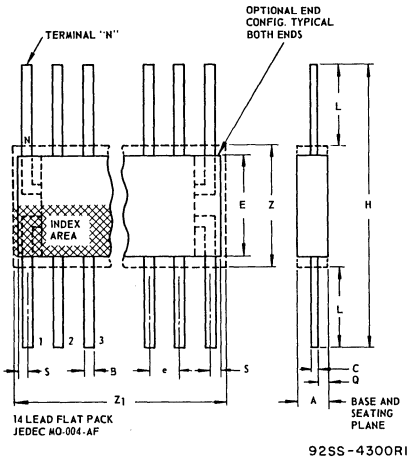
MODIFIED 16-LEAD QUAD-IN-LINE PLASTIC PACKAGE WITH INTEGRAL FLAT WING–TAB HEAT SINK



DIMENSIONS IN PARENTHESES ARE MILLIMETER EQUIVALENTS OF THE BASIC INCH DIMENSIONS

92CM-25045

14-LEAD CERAMIC FLAT PACKAGE JEDEC MO-004AF



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.008	0.100		0.21	2.54
B	0.015	0.019	1	0.381	0.482
C	0.003	0.006	1	0.077	0.152
e	0.050 TP		2	1.27 TP	
E	0.200	0.300		5.1	7.6
H	0.600	1.000		15.3	25.4
L	0.150	0.350		3.9	8.8
N	14		3	14	
Q	0.005	0.050		0.13	1.27
S	0.000	0.050		0.00	1.27
Z	0.300		4	7.62	
Z_1	0.400		4	10.16	

NOTES:

1. Refer to Rules for Dimensioning Peripheral Lead Outlines.
2. Leads within .005" (.12 mm) radius of True Position (TP) at maximum material condition.
3. N is the maximum quantity of lead positions.
4. Z and Z_1 determine a zone within which all body and lead irregularities lie.

MODIFIED 16-LEAD DUAL-IN-LINE PLASTIC PACKAGE WITH INTEGRAL HEAT SINK

